

Resat Orhun Konak

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Politecnico di Torino Computer Engineering graduate (BSc) and aspiring PoliMi Internet Engineering candidate. Focused on high-availability network design and Layer 3 switching fabrics. Currently bridging theoretical communication systems knowledge with hands-on CCNA-level implementation.

Education

Politecnico di Torino, BSc in Computer Engineering

September 2022 – March 2026

- **Special Focus: Network Infrastructure** (Protocols, Routing, and Switching), Signal Analysis and Processing, Computer Architecture, Applied Electronics
- Graduation Average Grade: 88/110

Experience

Student Ambassador, Reply – Torino

April 2024 – Present

- **Represented the company at events**, effectively bridging the communication gap between the organization and the technical student body.
- **Coordinated promotional and outreach campaigns**, demonstrating **organizational skills** and strict adherence to deadlines and project timelines. Supported company initiatives by managing assigned tasks autonomously and collaborating with diverse teams, ensuring successful execution of objectives.

Projects

Secure BGP Internet Edge & DMZ Architecture

- **Architected a dual-homed internet gateway implementing External BGP (eBGP)** peering with multiple upstream ISPs (AS 15169) under a local Autonomous System (AS 65000) to establish border gateway redundancy.
- Integrated protocol boundaries by configuring dynamic route redistribution between the eBGP edge and a Multi-Area **OSPF Area 0 backbone**, using floating static routes (Administrative Distance 10) to ensure automated link-failover.
- Deployed secure **network address translation** policies, combining **Port Address Translation (PAT Overload)** for internal LAN traffic with targeted Static NAT port forwarding to map public requests strictly to a hidden **DMZ Web Server (VLAN 200)** on TCP Port 80.
- **Engineered a zone-based security policy** by developing an Extended Access Control List (SECURE-DMZ) that leverages TCP flag matching via the established keyword to permit return traffic while blocking all unauthorized lateral movement into internal enterprise subnets.
- **Validated structural integrity** through real-time Cisco IOS verification states (show ip bgp summary, show ip access-lists), capturing incremental match metrics to prove active perimeter containment.

Redundant Enterprise Campus Network Architecture

- Engineered a high-availability “Triangle” Core utilizing three Cisco 3650 Layer 3 switches to eliminate single points of failure and ensure perfect backbone resilience.
- Manually tuned **Spanning Tree Protocol (STP)** priorities (4096, 8192, 12288) and aggregated bandwidth via **LACP (802.3ad) EtherChannels** for optimized, deterministic, and loop-free path selection.
- Configured **Inter-VLAN Routing** via **Switched Virtual Interfaces (SVIs)** and implemented **DHCP Relay** (ip helper-address) to provide dynamic IP allocation across segmented Student (VLAN 10), Staff (VLAN 20), and Management (VLAN 99) subnets.
- Hardened the Access Layer with **BPDUGuard** and **PortFast** on edge ports and applied **Root Guard** on core downlinks to protect the integrity of the spanning-tree topology.
- Established a **Point-to-Point /30 Transit Link** via “no switchport” routed interfaces to a Cisco ISR4321 Router, configuring **Static Routes** for end-to-end ICMP reachability to external WAN infrastructure.

Automated WAN Migration & Security Compliance - In progress

- Engineered a **Python-based automation** framework using **Netmiko** and YAML to deploy **Infrastructure-as-Code (IaC)** principles across an enterprise network topology.
- Architected an **Out-of-Band (OOB) management** plane using an air-gapped unmanaged switch, enabling secure, independent script execution and data-plane provisioning.
- Orchestrated the automated migration from a flat Layer 2 architecture to a routed Wide Area Network, dynamically provisioning OSPF transit links and inter-VLAN routing within Cisco Modeling Labs (CML).

Skills

- **Certification:** Cisco Certified Network Associate (CCNA) - In progress.
- **Networking & Security:** eBGP, Multi-Area OSPF, Route Redistribution, Floating Static Routes, Inter-VLAN Routing (SVIs), Layer 3 Switch Stacking, Router-on-a-Stick (802.1Q), EtherChannel (LACP), Static NAT, PAT Overload, Extended Access Control Lists (ACLs), TCP Flag Matching (established), RIPv2, STP Tuning (Root Bridge), BPDU Guard, Root Guard, PortFast, DHCP Relay.
- **Programming & Tools:** C, Java, Python, MATLAB, Cisco Packet Tracer, GitLab, Oracle SQL Server, Microsoft Office.
- **Automation & Environments:** Netmiko, YAML, Cisco Modeling Labs (CML).